Atadzhan Baratov

Principal AI/ML Engineer | Autonomous Systems • Multi-Agent Architectures • Long-Context RAG & AI Orchestration • Cloud Data & MLOps

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SUMMARY

Principal-level AI Engineer with over **13+ years of experience** designing, scaling, and optimizing machine learning, data, and cloud systems across **AI Research, FinTech, Cloud Services, and Public Sector** domains. Specializes in **AI infrastructure, MLOps automation, and multi-cloud architecture**, with a strong focus on **agentic and autonomous AI systems**—including multi-agent coordination, tool-driven execution, long-horizon reasoning, and agent-optimized retrieval (RAG). Delivers measurable gains in scalability, cost-efficiency, compliance, and operational reliability. Combines research innovation with production reliability and leads global teams in building **scalable, governed, and high-performance AI systems**.

Education

University of Northampton | Northampton, UK

MSc in Data Science and Artificial Intelligence , 09/2010 – 11/2011

BSc in Computer Science, 09/2007 – 06/2010

PROFESSIONAL EXPERIENCE

Principal AI Engineer, *OpenAI* 08/2025 – Present | Remote (US HQ)

- Designed and implemented **scalable, distributed AI infrastructure** across AWS, GCP, and Azure, improving GPU utilization by **~40%** for large-scale training and inference workloads.

- Led development of **modular ML platforms** integrating data ingestion, pipeline orchestration, and automated fine-tuning for foundation models, accelerating experimentation and deployment cycles.

- Built **advanced RAG and knowledge retrieval systems** using LangChain, Pinecone, and vector databases to support complex conversational and agentic workflows.

- Pioneered **LoRA and QLoRA fine-tuning pipelines**, reducing iteration cycles by **33%+** while maintaining model quality.

- Optimized **GPU cluster scheduling** using Kubernetes, Ray, and SLURM, improving fault tolerance, isolation, and availability.

- Developed **real-time observability frameworks** for drift, bias, and model health using Prometheus, Grafana, Arize AI, and Evidently.

- Implemented **enterprise-grade security and governance controls** (RBAC, encryption, secret management) for cross-cloud AI workloads.

- Delivered **£3M+ annual cost savings** through infrastructure optimization, scheduling efficiency, and workload orchestration.

- Extended AI infrastructure into **healthcare analytics, education systems, and enterprise automation** domains.

Senior Agentic AI Engineer, *OpenAI* 02/2022 – 08/2025| Remote (US HQ)

- Led design and implementation of **stateful agentic architectures** enabling long-horizon reasoning through **multi-agent coordination, tool execution, interrupt handling, and checkpoint-based recovery**.

- Built **graph-based agent controllers and decision state machines** supporting recursive planning, dynamic task decomposition, and adaptive execution under uncertainty.

- Architected **agent-optimized RAG systems**, including hierarchical chunking, hybrid retrieval, relevance scoring, and retrieval-timing strategies for long-context reasoning.

- Designed **context and memory frameworks** combining token-budget optimization, multi-turn pruning, and recursive memory consolidation to preserve coherence across extended agent lifecycles.

- Developed **evaluation and reliability platforms** for non-deterministic agents, including long-horizon failure simulation, regression tracking, and behavioral stability metrics.

- Productionized agentic workloads across **AWS, GCP, and Azure**, improving utilization by **40%+** via autoscaling and execution isolation.

- Embedded **governance, safety, and auditability** directly into agent pipelines (RBAC, lineage, GDPR-aligned controls).

- Authored internal architectural guidance on **agent design patterns, context efficiency, and evaluation strategies**.

Senior Cloud | MLOps Engineer, ***AWS*** 01/2018 – 01/2022| Hybrid (London, UK)

- Designed enterprise-grade **MLOps frameworks** using **AWS SageMaker**, **Terraform**, - **CloudFormation**, and **CodePipeline** for **Fortune 500 clients**.

- Reduced model **deployment time** from **three weeks to two days** by **containerising pipelines** and implementing **CI/CD automation**.

- Built scalable **AutoML services** with **XGBoost**, **LightGBM**, and **Scikit-learn**, driving **multi-million-pound compute savings**.

- Led **cross-region replication** and **encryption strategies** for ML workloads, achieving **SOC 2 Type II** and **ISO 27001 compliance**.

- Created unified **feature stores** and **metadata registries**, ensuring **model reproducibility** across **multiple ML teams**.

- Integrated **cost observability dashboards** using **AWS Lambda**, **Billing APIs**, **Athena**, and **QuickSight**, cutting **infrastructure costs by 25%+**.

- Partnered with **Professional Services** to deliver **AI-ready architectures** for **FinTech**, **logistics**, and **healthcare clients**.

- Delivered **keynote sessions** on **production ML pipelines** and **governance automation** at **AWS Summit London**.

- Mentored engineers in **Terraform best practices**, **data security**, and **MLOps automation principles**.

Data Engineer | ML Systems Developer, ***Revolut*** 04/2015 – 12/2017 | Onsite (London, UK)

- Engineered real-time **data pipelines** using **Kafka**, **Spark Streaming**, and **PostgreSQL**, processing **10M+ transactions daily**.

- Developed **fraud detection and credit risk models** using **TensorFlow**, **Keras**, and **Python**, significantly improving **model accuracy**.

- Built and maintained **feature stores** for **predictive modelling**, reducing **model delivery timelines by 50%**.

- Integrated **ML systems** into **customer-facing products** for **risk scoring**, **KYC**, and **AML automation**.

- Automated **A/B testing** and **experiment tracking** with **Apache Airflow** and **MLflow**, accelerating **product iteration**.

- Developed **monitoring dashboards** using **Grafana** and the **ELK Stack** to track **latency**, **drift**, and **system health**.

- Collaborated with **compliance teams** to align **AI pipelines** with **PSD2** and **GDPR regulations**.

- Recognized with the **Revolut Data Innovation Award** for **real-time fraud analytics architecture**.

Software Engineer, ***Capgemini UK*** 07/2012 – 04/2015 | Hybrid (Northampton, United Kingdom)

- Developed **RESTful APIs** and **enterprise microservices** using **Java**, **Spring Boot**, and **Oracle**.

- Automated **cloud migrations** to **AWS** using **Python** and **CloudFormation**, improving **delivery efficiency**.

- Created large-scale **ETL workflows** using **Informatica**, **SQL Server**, and **Python**, processing **terabytes of data nightly**.

- Introduced **CI/CD automation** via **Jenkins**, **Git**, and **Maven**, enhancing **deployment reliability**.

- Built **analytics dashboards** for **NHS clients** using **Power BI** and **Tableau**, reducing **manual reporting by 60%**.

- Led small **technical teams** delivering **secure data platforms** for **public-sector modernisation projects**.

ML Intern, ***Microsoft Research Cambridge,*** **06/2011 – 06/2012 | Remote(Cambridge, UK)**

- Conducted applied **computer vision research** in **object detection** and **deep learning** using **Python** and **OpenCV**.

- Built **GPU-accelerated training pipelines** with **CUDA** and **PyTorch**, dramatically speeding up **experimentation**.

- Developed early **convolutional neural network prototypes** for **image classification tasks**.

- Co-authored internal **research reports** on **feature pyramid extraction** and **visual representation learning**.

TECHNICAL SKILLS

Programming Languages
Python, Java, Go, Scala, SQL, Bash, JavaScript, R

Agentic AI & Autonomous Systems
Multi-agent systems, agent state machines, decision graphs, tool-driven execution, interrupt/resume logic, long-horizon reasoning, agent memory & context management, agent-optimized RAG, evaluation of non-deterministic behavior, human-in-the-loop systems

Machine Learning & Generative AI
PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Ray, LangChain, MLflow, Weights & Biases, SHAP, LIME
Multimodal & Foundation Models: CLIP, Flamingo, ViLT, VisualBERT, DALL·E, Stable Diffusion, Perceiver IO, GANs
Embeddings & Representation Learning: multimodal embeddings, vector similarity search, representation fusion
Speech & Audio AI: speech recognition, text-to-speech (Google TTS, OpenAI TTS, Coqui)

Data Engineering & Retrieval Systems
Apache Spark, Airflow, dbt, Kafka, Flink, Snowflake, Redshift, PostgreSQL, ElasticSearch, Hadoop
Vector & Knowledge Stores: Pinecone, Weaviate, FAISS, Neo4j

MLOps, AI Platforms & Infrastructure
AWS (SageMaker, Lambda, S3, EC2, EKS, Bedrock),
GCP (Vertex AI, BigQuery, multimodal endpoints),
Azure (Machine Learning Studio)
Kubernetes, Docker, Terraform, ArgoCD, Helm, Jenkins, GitHub Actions
Model Serving: BentoML, TorchServe, Triton Inference Server

Monitoring, Evaluation & Observability
Prometheus, Grafana, Arize AI, Evidently, Datadog, New Relic, ELK Stack
Model performance monitoring, drift detection, bias analysis, cost and latency tracking

Cloud, DevOps & Distributed Systems
CloudFormation, IAM, RBAC, Secrets Management, Load Balancing, Auto Scaling, VPC Networking
Serverless AI deployments, distributed scheduling, edge AI deployment

Data Governance, Security & Compliance
Apache Atlas, data lineage, encryption policies, access control, GDPR, SOC 2, ISO 27001

Visualization & Analytics

Power BI, Tableau, QuickSight, Matplotlib, Seaborn, Plotly, Dash

APIs, Frameworks & Engineering Practices

FastAPI, REST, GraphQL, Git, CI/CD automation, Python packaging

Developer Tools: Jupyter, Streamlit, Gradio

Ways of Working: Agile, Scrum, Jira, Confluence